

EXPERIENCE WITH A BUNCH LENGTHENING CAVITY AT THE APS*

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Abstract

The Advanced Photon Source Upgrade (APS-U) at Argonne National Laboratory (ANL) provides hard x-ray photon beams at 6 GeV with a brightness 500 times greater than the original machine. A bunch lengthening cavity is used to decrease the effects of Touschek scattering (on beam lifetime) and intrabeam scattering (on beam emittance). The superconducting RF (SRF) cavity operates at 2 K in a passive, i.e. beam-driven mode at the 4th harmonic (1408 MHz) of the main RF system. A helium cryoplant provides 2 K refrigeration for the SRF cavity.

CAVITY AND CRYOMODULE DESIGN

In 2014 a concept was developed for a superconducting higher harmonic cavity (HHC) for the APS-U storage ring. The design, assembly, and testing of the device were conducted in the ANL Physics Division [1]. Scope includes:

- A single 1.4 GHz elliptical-cell SRF cavity.
- A low-level RF (LLRF) system to provide RF diagnostics and control cavity amplitude via a slow tuner.
- A cryomodule to support the SRF cavity and associated RF, cryogenic, and beam vacuum subsystems.
- A liquid helium refrigerator providing up to 50 W of cooling power at 2.1 K.
- Height and width consistent with tunnel dimensions.
- Overall cryomodule length not to exceed half of an APS-U 5-meter straight section.

Cavity and Cryomodule

Figure 1 shows the completed SRF cavity/liquid helium (LHe) tank assembly. Symmetrically placed variable couplers (ports shown on the right side) have not proven necessary in operation. The bellows on the left provides a flexible element in the LHe tank for cavity tuning.



Figure 1: LHe tank containing SRF cavity.



Figure 2: Cryomodule lid with suspended cold mass.

The cold mass assembly is suspended from the lid of the module (see Fig. 2). The angled end-walls of the lower vessel (see Fig. 3 below) allow the lid assembly, including vacuum gate valves at each end, to be lowered into the vessel. The valve actuators emerge through the openings in the ends and seal against the inside walls of the openings.

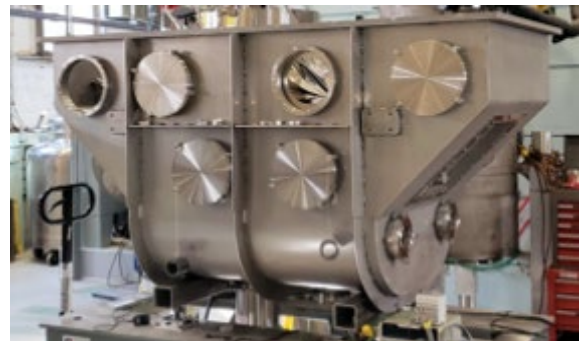


Figure 3: Cryomodule lower vacuum vessel.

RF Systems

Figure 4 provides an overview of the main RF systems and the HHC. The electron beam induces voltage in the HHC to lengthen the beam bunches while the main 352 MHz systems replenish beam energy which is lost to synchrotron radiation. The HHC field (see detuning curve, Fig. 5) depends on beam current, external Q, and amount of detuning. Cavity parameters are listed in Table 1.

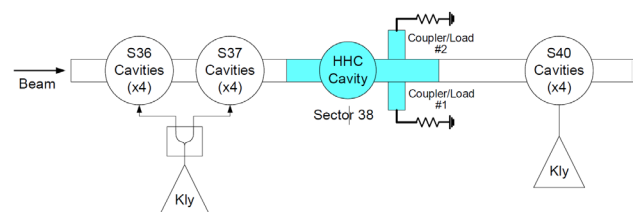


Figure 4: Main and harmonic RF systems.

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Table 1: Cavity Parameters

Parameter	Symbol	Unit	Value
Operating Temperature	T	K	4.5
R/Q	r/Q	Ohm	109
Cavity Quality Factor	Q_0		2×10^8
External Q	Q_{ext}		2×10^5 - 2×10^7
Cavity Loaded Bandwidth	Δf_{BW}	kHz	5.4
Detuning Frequency		kHz	15
Cavity Resonant Frequency	f_r	MHz	1408
Beam-Induced Voltage	V_b	MV	0.90
Detuning angle	ψ_h	degrees	78.9
Beam Loss Power	P_b	kW	32
Cavity Wall Loss Power	P_{wall}	W	35
Peak Surface Electric Field	E_{peak}	MV/m	17
Peak Surface Magnetic Field	B_{peak}	mT	35

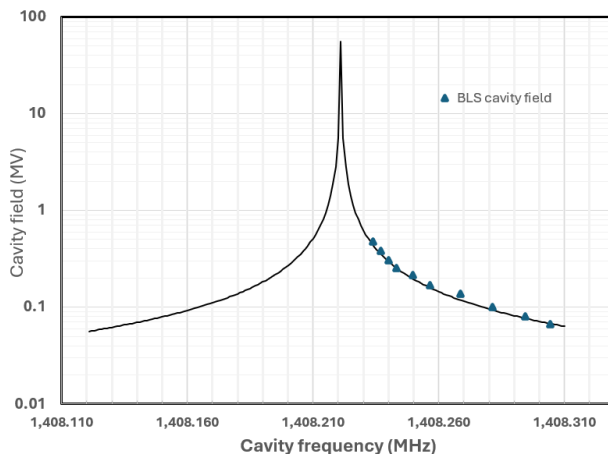


Figure 5: BLS HHC resonance detuning curve.

When bunch lengthening is not desired, the cavity may be “parked” off-resonance which positions the cavity in a safe state when beam is still present in the storage ring. Parking mode is a critical feature of the cavity as beam cannot be safely circulated if the HHC is in an uncontrolled frequency state.

INSTALLATION

The cryoplant arrived in January of 2021. Vendor commissioning and acceptance took place in the fall of 2021, after which the plant was shut down. This phase of commissioning involved only the Linde LR140 refrigerator/liquefier, its associated recycle compressor, and the 1000-Liter LHe storage dewar together with ancillary cryogenic piping.

The remaining BLS hardware components were prepared in the interim between cryoplant commissioning and system restart, which occurred in late 2023. Major installation efforts during the interim included the cryogenic distribution system (CDS) and the HHC cryomodule itself. Figure 6 shows the HHC ready to be moved into its location in the empty storage ring tunnel.



Figure 6: HHC transport to tunnel location.

COMMISSIONING

Cryogenic Systems

Operational commissioning of the complete BLS system including LHe refrigerator, LHe and liquid nitrogen (LN2) distribution systems, and the HHC cryomodule began in December of 2023. The system was brought on-line in the following stages:

- Cooldown and 4.2 K operation of the LHe refrigerator (including internal LN2 shield) and 1000 L LHe storage dewar.
- Cooldown and 80K operation of the LN2 shields for the distribution system and HHC.
- Cooldown and 4.2 K operation of the distribution system and HHC.
- Initiation of the room-temperature helium vacuum pump (HVP) systems for sub-atmospheric operation of the HHC at 2.06 K.

Initial operation was uneventful. The system reached 4.2 K in about 18 hours and maintained a stable liquid level at the 150 L set point, controlled by a heater in the 1000 L dewar. After two weeks, the HVP system were engaged and reached a stable HHC LHe bath pressure of 37 mbar. The pressure at the HVP system inlet was 33 mbar, in line with the expected pressure drop through the He gas warmup heat exchanger located between the LHe bath and the HVP system inlet.

Several issues were encountered during the commissioning phase. Those listed below were corrected except for the LN2 distribution system (LNDS), which requires implementation of a facility-wide solution.

- LN2 delivery pressure fluctuations due variable demand on the facility-wide LNDS
- LHe supply valve regulation due to actuator issues
- Significant frost spots on the exterior of the He gas warmup heat exchanger due to lack of circulation in the heated water bath of the exchanger.

HHC and Cryomodule Systems

The HHC and cryomodule systems were commissioned with beam during dedicated machine intervention periods.

These opportunities initially took place during APS-U commissioning and have continued sporadically through the course of subsequent operational runs. Initial commissioning using the standard lattice (162 electron bunches or greater) reached the desired beam current of 200 mA with a cavity voltage of 700 kV. Voltage later reached between 1.1 and 1.2 MV. Figures 7 and 8 show the commissioning history from 15 December 2024 through 16 April 2025. Commissioning in 48-bunch timing mode has also reached acceptable performance levels, and further studies are planned.

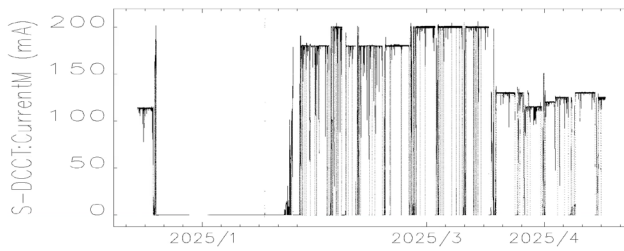


Figure 7: Beam current (mA) vs. time.

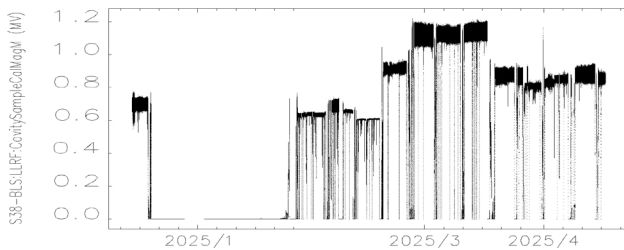


Figure 8: HHC voltage (MV) vs. time.

OPERATION

In February 2025 the cavity demonstrated a stable 1.1 MV accelerating gradient with 200 mA beam for user operation. Figures. 9 and 10 show a six-hour plot of beam current and cavity voltage during timing mode operation on 15 April 2025. Figures. 11 and 12 show a similar plot during regular lattice operation over a one-hour period on 15 March 2025. Beam bunch decay and beam swap-out is evident in Figs. 9 and 11.

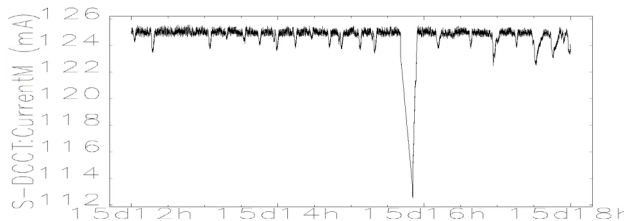


Figure 9: Beam current (mA) vs. time.

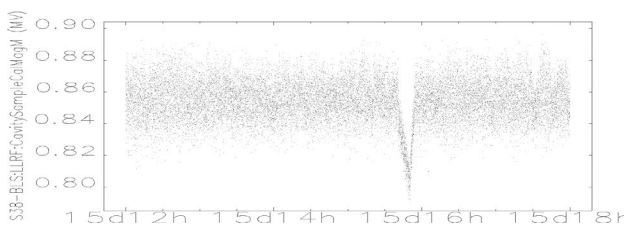


Figure 10: HHC voltage (MV) vs. time.

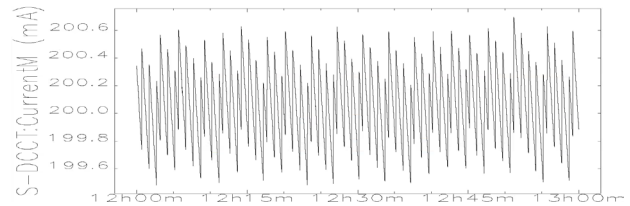


Figure 11: Beam current (mA) vs. time.

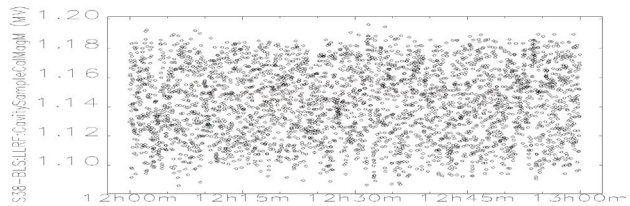


Figure 12: HHC voltage (MV) vs. time.

The HHC helium bath initially regulated at a pressure of 37 +/- 2 mbar (see Fig. 13). Although the cavity was able to run stably with this level of pressure fluctuation, a tighter range is desirable to improve overall cryoplant operation. Although bath pressure is regulated by a butterfly valve at the HVP system inlet, a separate JT expansion valve regulates the liquid level of the helium bath. When the JT valve's mechanical actuator was replaced with a more precise pneumatic unit in May 2025, stability improved to +/- 0.4 mbar (see Fig. 14).

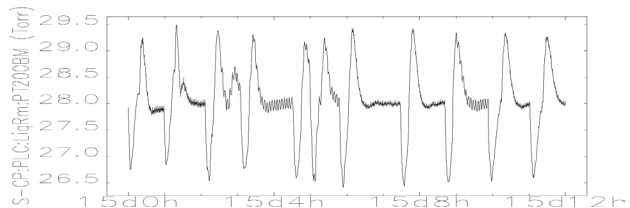


Figure 13: Pressure stability with mechanical actuator

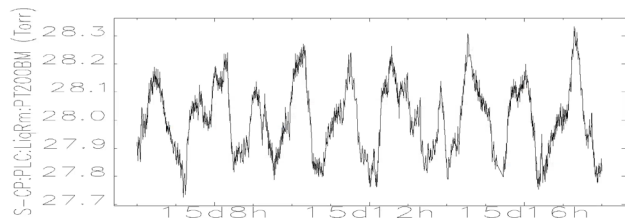


Figure 14: Pressure stability with pneumatic actuator

CONCLUSION

The APS bunch lengthening system (BLS) improves beam lifetime by a factor of 2-4 depending on the mode of operation. A helium cryoplant and ancillary systems were installed to provide 2 K refrigeration for the srf cavity. In February 2025 the BLS demonstrated a stable 1.1 MV with 200 mA beam for user operation.

REFERENCES

- [1] M. P. Kelly *et al.*, "Superconducting harmonic cavity for bunch lengthening in the APS Upgrade", in *Proc. SRF'19*, Dresden, Germany, June-July 2019, pp. 715-717.
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